

Exercise 4 – Solution Proposal

Charge-transfer kinetics: Butler–Volmer

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Introduction

In this document, we investigate charge-transfer kinetics using the Butler–Volmer equation. The questions and notation are based on the problem sheet [1]. In the previous exercises, we considered equilibrium thermodynamics and electric-double-layer structure. Here, we focus on the kinetic relation between overpotential and faradaic current.

The aim is to understand how the current density depends on the overpotential, the exchange current density, the transfer coefficient, the number of transferred electrons, and the temperature. We also compare the full Butler–Volmer equation with its anodic and cathodic Tafel approximations.

Equations: Overpotential and charge-transfer reaction

The overpotential is defined as the deviation of the electrode potential from its equilibrium value,

$$\eta = E - E_{\text{eq}}. \quad (1)$$

A generic electrochemical charge-transfer reaction can be written as



At equilibrium,

$$\eta = 0, \quad i = 0. \quad (3)$$

The net current vanishes because the anodic and cathodic reaction currents cancel each other.

Equations: Butler–Volmer equation

The Butler–Volmer equation relates the net current density to the overpotential:

$$i(\eta) = i_0 \left[\exp \left(\frac{(1 - \alpha)nF\eta}{RT} \right) - \exp \left(-\frac{\alpha nF\eta}{RT} \right) \right]. \quad (4)$$

Here, i_0 is the exchange current density, α is the cathodic charge-transfer coefficient, $1 - \alpha$ is the anodic charge-transfer coefficient, n is the number of transferred electrons, F is the Faraday constant, R is the gas constant, and T is the temperature.

The exchange current density i_0 sets the current scale. The transfer coefficient α controls the asymmetry between the anodic and cathodic branches.

Equations: Anodic and cathodic current densities

The Butler–Volmer current can be written as the sum of an anodic and a cathodic contribution:

$$i(\eta) = i_a(\eta) + i_c(\eta). \quad (5)$$

The anodic current density is

$$i_a(\eta) = i_0 \exp \left(\frac{(1 - \alpha)nF\eta}{RT} \right), \quad (6)$$

and the cathodic current density is

$$i_c(\eta) = -i_0 \exp \left(-\frac{\alpha nF\eta}{RT} \right). \quad (7)$$

At equilibrium, $\eta = 0$, these two contributions have equal magnitude and opposite sign:

$$i_a(0) = i_0, \quad (8)$$

$$i_c(0) = -i_0, \quad (9)$$

$$i(0) = i_a(0) + i_c(0) = 0. \quad (10)$$

Thus, i_0 is not a net current at equilibrium. It is the magnitude of the forward and backward exchange currents at equilibrium.

Equations: Tafel approximations

For sufficiently large positive overpotential, the anodic exponential dominates:

$$\eta \gg 0 \implies i(\eta) \approx i_a(\eta) = i_0 \exp\left(\frac{(1-\alpha)nF\eta}{RT}\right). \quad (11)$$

For sufficiently large negative overpotential, the cathodic exponential dominates:

$$\eta \ll 0 \implies i(\eta) \approx i_c(\eta) = -i_0 \exp\left(-\frac{\alpha nF\eta}{RT}\right). \quad (12)$$

Taking the logarithm gives the anodic Tafel form

$$\eta = \frac{RT}{(1-\alpha)nF} \ln\left(\frac{i}{i_0}\right), \quad (13)$$

and the cathodic Tafel form

$$\eta = -\frac{RT}{\alpha nF} \ln\left(\frac{|i|}{i_0}\right). \quad (14)$$

Thus, in a Tafel plot of η against $\ln(|i|)$, the large-overpotential branches become approximately linear.

Equations: Base-10 Tafel slopes

Experimental Tafel plots are often written using $\log_{10}(|i|)$ rather than $\ln(|i|)$. The corresponding base-10 Tafel slopes are

$$b_{a,10} = \frac{2.303RT}{(1-\alpha)nF}, \quad (15)$$

$$b_{c,10} = -\frac{2.303RT}{\alpha nF}. \quad (16)$$

These slopes have units of voltage per decade. Increasing T increases the magnitude of the Tafel slope, while increasing n decreases the magnitude of the Tafel slope. Changing α changes the anodic and cathodic slopes in opposite ways.

Table 1: Fundamental constants

Symbol	Value	Unit
N_A	6.022×10^{23}	mol^{-1}
k_B	1.381×10^{-23}	J K^{-1}
e_0	1.602×10^{-19}	C
F	96485	C mol^{-1}
R	8.314	$\text{J mol}^{-1} \text{K}^{-1}$

Values are given in SI units.

Table 2: Model parameters and variables in the Butler–Volmer equation

Symbol	Unit	Meaning
η	V	Overpotential, i.e. deviation from the equilibrium potential.
E	V	Electrode potential.
E_{eq}	V	Equilibrium electrode potential.
i	A m^{-2}	Net faradaic current density.
i_a	A m^{-2}	Anodic current density, corresponding to oxidation.
i_c	A m^{-2}	Cathodic current density, corresponding to reduction.
i_0	A m^{-2}	Exchange current density; magnitude of the anodic and cathodic currents at equilibrium.
α	—	Cathodic charge-transfer coefficient.
$1 - \alpha$	—	Anodic charge-transfer coefficient.
n	—	Number of transferred electrons.
T	K	Temperature.
$b_{a,10}$	V decade^{-1}	Anodic base-10 Tafel slope.
$b_{c,10}$	V decade^{-1}	Cathodic base-10 Tafel slope.

1 Model setup: Butler–Volmer equation

Question: 1 Butler–Volmer equation

For the Butler–Volmer equation, please answer the following questions:

- 1.1) What is the physical meaning of the overpotential η ?
- 1.2) Why is the net current density zero at $\eta = 0$ even though charge transfer still occurs?
- 1.3) What is the meaning of the exchange current density i_0 ?
- 1.4) How does increasing i_0 change the Butler–Volmer curve?
- 1.5) What is the role of the transfer coefficient α ?
- 1.6) What changes when $\alpha = 0.5$ is replaced by $\alpha \neq 0.5$?

Solution: 1 Butler–Volmer equation

We use the Butler–Volmer equation and the anodic–cathodic decomposition introduced in the Introduction:

$$i(\eta) = i_a(\eta) + i_c(\eta). \quad (17)$$

Here, i_a is the anodic oxidation current and i_c is the cathodic reduction current. With the sign convention used in this exercise,

$$i_a(\eta) > 0, \quad i_c(\eta) < 0. \quad (18)$$

The questions below focus on the physical interpretation of the parameters η , i_0 , and α .

1.1) Physical meaning of the overpotential η

The overpotential is the deviation of the electrode potential from its equilibrium value:

$$\eta = E - E_{\text{eq}}. \quad (19)$$

At equilibrium,

$$\eta = 0. \quad (20)$$

If $\eta > 0$, the electrode potential is shifted in the positive direction relative to equilibrium. This favors the anodic direction, so oxidation becomes dominant and the net current becomes positive. If $\eta < 0$, the electrode potential is shifted in the negative direction relative to equilibrium. This favors the cathodic direction, so reduction becomes dominant and the net current becomes negative.

Thus, for the sign convention used here,

$$\boxed{\eta > 0 \Rightarrow i > 0, \quad \eta < 0 \Rightarrow i < 0.}$$

The overpotential therefore measures how strongly the electrode is driven away from equilibrium. A larger absolute value $|\eta|$ corresponds to a stronger kinetic driving force for charge transfer.

1.2) Net current at equilibrium

At equilibrium, the overpotential is zero:

$$\eta = 0. \quad (21)$$

From the anodic and cathodic current expressions introduced in the Introduction, the two branch currents become

$$i_a(0) = i_0, \quad (22)$$

$$i_c(0) = -i_0. \quad (23)$$

Therefore, the net current density is

$$i(0) = i_a(0) + i_c(0) = i_0 - i_0 = 0. \quad (24)$$

Hence,

$$i(0) = 0$$

even though charge transfer still occurs microscopically.

The reason is that, at equilibrium, the anodic and cathodic reaction currents have equal magnitude and opposite sign. Forward and backward charge-transfer events still occur, but they cancel in the net current.

1.3) Meaning of the exchange current density i_0

The exchange current density i_0 is the magnitude of the anodic and cathodic current densities at equilibrium:

$$i_a(0) = i_0, \quad i_c(0) = -i_0. \quad (25)$$

Thus, i_0 is not a net current at equilibrium. Instead, it measures how rapidly charge transfer occurs in both directions when the system is at equilibrium.

A large exchange current density means that the electrode reaction is kinetically fast. The forward and backward reactions both occur rapidly, even though their net difference is zero at equilibrium. A small exchange current density means that the electrode reaction is kinetically slow.

Thus,

$$i_0 \text{ measures the kinetic activity of the electrode reaction at equilibrium.}$$

1.4) Effect of increasing i_0

In the Butler–Volmer equation, the exchange current density i_0 multiplies both the anodic and cathodic current contributions. Therefore, increasing i_0 increases the magnitude of both branch currents.

If i_0 is multiplied by a constant factor, the whole Butler–Volmer current is multiplied by the same factor:

$$i_0 \longrightarrow ai_0 \implies i(\eta) \longrightarrow ai(\eta). \quad (26)$$

Thus, increasing i_0 changes the vertical scale of the Butler–Volmer curve, but it does not change the basic shape controlled by α , n , and T .

Hence,

$$i_0 \uparrow \implies |i(\eta)| \uparrow$$

for all nonzero overpotentials.

A reaction with large i_0 therefore needs only a small overpotential to produce a large current. A reaction with small i_0 requires a larger overpotential to produce the same current density.

This behavior is also visible in the bottom row of Figure 1, where increasing i_0 increases the current magnitude without changing the qualitative structure of the anodic, cathodic, and net-current curves.

1.5) Role of the transfer coefficient α

The transfer coefficient α controls how the overpotential is distributed between the cathodic and anodic activation barriers.

In the notation of the Introduction, the cathodic branch contains the factor α , while the anodic branch contains the factor $1 - \alpha$. Therefore, α determines how sensitively the cathodic branch responds to negative overpotential, while $1 - \alpha$ determines how sensitively the anodic branch responds to positive overpotential.

Hence,

$$\alpha \text{ controls the asymmetry between the cathodic and anodic branches.}$$

If α is increased, the cathodic branch becomes more sensitive to negative overpotential, while the anodic branch becomes less sensitive to positive overpotential. If α is decreased, the anodic branch becomes more sensitive to positive overpotential, while the cathodic branch becomes less sensitive to negative overpotential.

1.6) Symmetric and asymmetric Butler–Volmer curves

For the symmetric case,

$$\alpha = 0.5, \tag{27}$$

we have

$$1 - \alpha = \alpha = 0.5. \tag{28}$$

Therefore, the anodic and cathodic exponential factors have the same absolute strength. The positive and negative branches of the Butler–Volmer curve are mirror images, up to the sign of the current.

In this case,

$$\alpha = 0.5 \implies \text{symmetric anodic and cathodic response.}$$

If $\alpha \neq 0.5$, the anodic and cathodic transfer coefficients are no longer equal:

$$1 - \alpha \neq \alpha. \tag{29}$$

One branch then grows more rapidly than the other. If $\alpha > 0.5$, the cathodic branch is more sensitive to negative overpotential, whereas the anodic branch is less sensitive to positive overpotential. If $\alpha < 0.5$, the opposite trend occurs.

Thus,

$$\alpha \neq 0.5 \implies \text{asymmetric Butler–Volmer curve.}$$

This behavior is visible in Figure 1. The top row shows an α -sweep and illustrates how changing α modifies the asymmetry between the anodic and cathodic branches. The bottom row shows an i_0 -sweep and illustrates how changing i_0 changes the current scale.

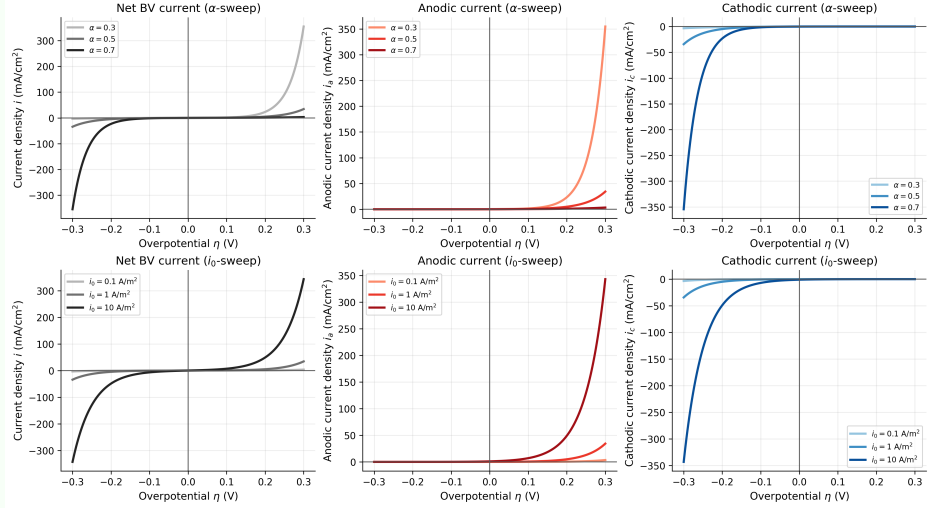


Figure 1: Butler–Volmer current decomposition for different transfer coefficients α and exchange current densities i_0 . The top row shows an α -sweep and compares the net Butler–Volmer current, anodic current, and cathodic current. The bottom row shows an i_0 -sweep and illustrates how increasing i_0 increases the current magnitude while preserving the qualitative shape of the response.

2 Tafel approximation

Question: 2 Tafel approximation

For the Tafel approximation, please answer the following questions:

- 2.1) Under which condition can the anodic Tafel approximation be used?
- 2.2) Under which condition can the cathodic Tafel approximation be used?
- 2.3) Why does the Butler–Volmer curve become approximately linear in a Tafel plot?
- 2.4) How does changing α affect the anodic and cathodic Tafel slopes?
- 2.5) How does changing T affect the Tafel slope?
- 2.6) How does changing n affect the Tafel slope?

Solution: 2 Tafel approximation

We use the Tafel approximations and base-10 Tafel slopes introduced in the Introduction. The key idea is that the full Butler–Volmer equation contains two exponential contributions. At sufficiently large positive or negative overpotential, one of these exponentials dominates the other.

2.1) Condition for the anodic Tafel approximation

The anodic Tafel approximation is used at sufficiently large positive overpotential:

$$\eta \gg 0. \quad (30)$$

In this regime, the anodic exponential dominates the Butler–Volmer equation. The cathodic contribution becomes small compared with the anodic contribution. Therefore,

$$i(\eta) \approx i_a(\eta). \quad (31)$$

Using the expression from the Introduction,

$$i(\eta) \approx i_0 \exp\left(\frac{(1-\alpha)nF\eta}{RT}\right). \quad (32)$$

Thus,

$$\boxed{\eta \gg 0 \implies \text{anodic Tafel approximation}}$$

Physically, this means that oxidation dominates and the reduction current is negligible.

2.2) Condition for the cathodic Tafel approximation

The cathodic Tafel approximation is used at sufficiently large negative overpotential:

$$\eta \ll 0. \quad (33)$$

In this regime, the cathodic exponential dominates the Butler–Volmer equation. The anodic contribution becomes small compared with the cathodic contribution.

Therefore,

$$i(\eta) \approx i_c(\eta). \quad (34)$$

Using the expression from the Introduction,

$$i(\eta) \approx -i_0 \exp\left(-\frac{\alpha n F \eta}{RT}\right). \quad (35)$$

Thus,

$$\boxed{\eta \ll 0 \implies \text{cathodic Tafel approximation}}$$

Physically, this means that reduction dominates and the oxidation current is negligible.

2.3) Linearity in a Tafel plot

In the anodic Tafel regime, the current is approximately

$$i \approx i_0 \exp\left(\frac{(1 - \alpha)n F \eta}{RT}\right). \quad (36)$$

Taking the logarithm gives

$$\eta = \frac{RT}{(1 - \alpha)n F} \ln\left(\frac{i}{i_0}\right). \quad (37)$$

This is linear in $\ln(i)$.

In the cathodic Tafel regime, the current is negative, so we use the magnitude $|i|$:

$$|i| \approx i_0 \exp\left(-\frac{\alpha n F \eta}{RT}\right). \quad (38)$$

Taking the logarithm gives

$$\eta = -\frac{RT}{\alpha n F} \ln\left(\frac{|i|}{i_0}\right). \quad (39)$$

This is linear in $\ln(|i|)$.

Thus,

$$\boxed{\text{The Tafel plot is linear because the logarithm of an exponential is linear.}}$$

The full Butler–Volmer curve is not linear everywhere in a Tafel plot. It is only approximately linear in the large-overpotential regions where one exponential dominates.

2.4) Effect of changing α on the Tafel slopes

Using base-10 logarithms, the anodic and cathodic Tafel slopes are

$$b_{a,10} = \frac{2.303RT}{(1 - \alpha)nF}, \quad (40)$$

$$b_{c,10} = -\frac{2.303RT}{\alpha nF}. \quad (41)$$

The anodic slope depends on $1 - \alpha$, whereas the cathodic slope depends on α .

If α increases, then $1 - \alpha$ decreases. Therefore, the anodic slope $b_{a,10}$ increases in magnitude:

$$\alpha \uparrow \implies 1 - \alpha \downarrow \implies b_{a,10} \uparrow. \quad (42)$$

At the same time, the magnitude of the cathodic slope decreases:

$$\alpha \uparrow \implies |b_{c,10}| \downarrow. \quad (43)$$

Thus,

$$\boxed{\alpha \text{ changes the anodic and cathodic Tafel slopes in opposite ways.}}$$

For $\alpha = 0.5$, the two branches have the same absolute slope:

$$b_{a,10} = -b_{c,10}. \quad (44)$$

For $\alpha \neq 0.5$, the absolute slopes are different.

2.5) Effect of changing temperature T

Both base-10 Tafel slopes are proportional to the temperature:

$$b_{a,10} = \frac{2.303RT}{(1 - \alpha)nF}, \quad (45)$$

$$b_{c,10} = -\frac{2.303RT}{\alpha nF}. \quad (46)$$

Therefore, increasing T increases the magnitude of both Tafel slopes:

$$T \uparrow \implies |b_{a,10}| \uparrow, \quad |b_{c,10}| \uparrow. \quad (47)$$

Hence,

$$\boxed{T \uparrow \implies \text{larger Tafel slope magnitude}}$$

In a Tafel plot of η against $\log_{10}(|i|)$, a larger slope means that a larger change in overpotential is needed to change the current by one decade. Equivalently, the current changes less steeply with overpotential.

2.6) Effect of changing the number of transferred electrons n

Both base-10 Tafel slopes are inversely proportional to the number of transferred electrons:

$$b_{a,10} = \frac{2.303RT}{(1 - \alpha)nF}, \quad (48)$$

$$b_{c,10} = -\frac{2.303RT}{\alpha nF}. \quad (49)$$

Therefore, increasing n decreases the magnitude of both Tafel slopes:

$$n \uparrow \implies |b_{a,10}| \downarrow, \quad |b_{c,10}| \downarrow. \quad (50)$$

Hence,

$$\boxed{n \uparrow \implies \text{smaller Tafel slope magnitude}}$$

In a Tafel plot of η against $\log_{10}(|i|)$, a smaller slope means that the same decade change in current requires a smaller change in overpotential. Thus, increasing n makes the current response steeper as a function of overpotential.

The effects of α , n , and T are illustrated in Figure 2. The left panel shows how changing α changes the anodic and cathodic slopes in opposite ways. The middle panel shows that increasing n decreases the slope magnitude. The right panel shows that increasing T increases the slope magnitude.

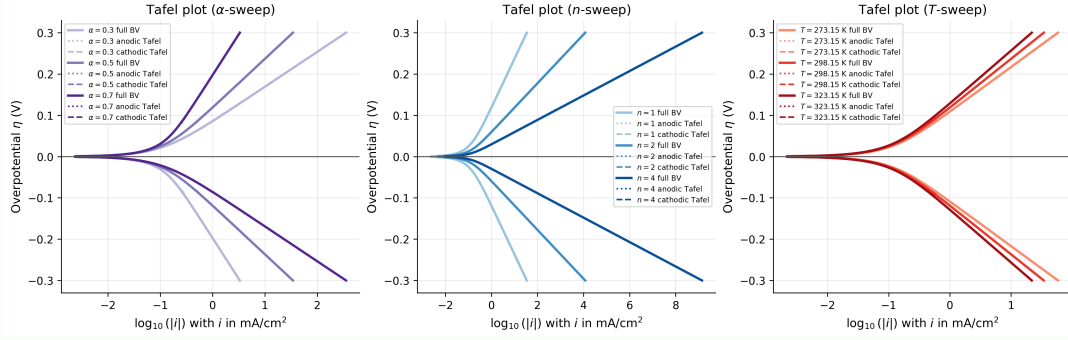


Figure 2: Conventional Tafel plots for different transfer coefficients α , numbers of transferred electrons n , and temperatures T . The full Butler-Volmer curve is compared with the anodic and cathodic Tafel approximations. The approximately linear regions correspond to the large-overpotential Tafel regimes.

3 Overall model assessment

Question: 3 Overall model assessment

For the Butler–Volmer equation, please answer the following questions:

- 3.1) What are the strengths of the Butler–Volmer equation?
- 3.2) What are the main limitations of the Butler–Volmer equation?
- 3.3) Why is Butler–Volmer not a complete mass-transport model?
- 3.4) Which parameters control the kinetic response of the electrode?
- 3.5) How does Butler–Volmer connect to the Nernst equation?
- 3.6) How does Butler–Volmer connect to electric-double-layer models?

Solution: 3 Overall model assessment

The Butler–Volmer equation is a compact kinetic model for faradaic charge transfer. It describes how the net current density changes when the electrode is driven away from equilibrium by an overpotential η .

In the notation used throughout this exercise,

$$i(\eta) = i_a(\eta) + i_c(\eta), \quad (51)$$

where the anodic branch describes oxidation and the cathodic branch describes reduction.

3.1) Strengths of the Butler–Volmer equation

The main strength of the Butler–Volmer equation is that it gives a simple relation between overpotential and faradaic current density. It captures the competition between anodic and cathodic charge-transfer reactions in one equation.

At equilibrium, the two branch currents cancel:

$$i_a(0) = i_0, \quad i_c(0) = -i_0, \quad i(0) = 0. \quad (52)$$

Thus, the model correctly predicts that the net current vanishes at equilibrium, even though microscopic charge transfer still occurs in both directions.

The Butler–Volmer equation also predicts the exponential growth of current away from equilibrium. For positive overpotential, the anodic branch dominates; for negative overpotential, the cathodic branch dominates. At sufficiently large overpotential, the model reduces to the anodic or cathodic Tafel approximation.

The main strengths are:

- it gives zero net current at equilibrium,
- it describes anodic and cathodic directions in one equation,
- it predicts exponential current growth with overpotential,
- it introduces i_0 as a measure of kinetic activity,
- it uses α to describe anodic–cathodic asymmetry,
- it naturally connects to Tafel analysis at large overpotential.

Thus,

Butler–Volmer is a compact model for charge-transfer kinetics.

3.2) Main limitations of the Butler–Volmer equation

The basic Butler–Volmer equation is a kinetic boundary relation. It describes the local charge-transfer rate at an electrode surface, but it does not describe all physical processes that may occur in an electrochemical cell.

In particular, the basic model does not explicitly include mass transport, concentration gradients, electric-double-layer structure, or changes in surface state. It assumes that the parameters i_0 and α are already known and remain fixed.

Important limitations are:

- it does not explicitly describe diffusion or convection,
- it does not include concentration gradients near the electrode,
- it does not describe the electric-double-layer potential profile,
- it treats i_0 and α as prescribed parameters,
- it does not include specific adsorption or surface restructuring,
- it does not include transport-limited current plateaus.

Thus,

Butler–Volmer is useful for kinetics, but it is not a complete interface model.

3.3) Why Butler–Volmer is not a complete mass-transport model

Mass transport describes how reactants and products move through the electrolyte. This may involve diffusion, migration, and convection. A complete mass-transport model would therefore require concentration fields, fluxes, and transport equations in the electrolyte.

The basic Butler–Volmer equation does not contain such transport equations. It relates the local current density to the local overpotential, but it does not determine how the reactant concentration changes with distance from the electrode.

For example, if a reactant is consumed rapidly at the surface, its concentration near the electrode may become smaller than its bulk concentration. This concentration depletion can limit the current. The basic Butler–Volmer equation does not predict this depletion by itself.

Thus,

Butler–Volmer is a charge-transfer relation, not a transport model.

To describe mass transport, Butler–Volmer must be coupled to additional equations, such as diffusion or Nernst–Planck transport equations.

3.4) Parameters controlling the kinetic response

The kinetic response in the Butler–Volmer equation is controlled mainly by i_0 , α , n , and T .

The exchange current density i_0 sets the current scale. Increasing i_0 increases the magnitude of both anodic and cathodic currents. Therefore, a larger i_0 corresponds to faster electrode kinetics.

The transfer coefficient α controls the asymmetry between the cathodic and anodic branches. The cathodic branch contains the factor α , whereas the anodic branch contains the factor $1 - \alpha$. Therefore, changing α changes the relative steepness of the two branches.

The number of transferred electrons n controls how strongly the current responds to the overpotential. A larger n makes the exponential factors more sensitive to η and decreases the magnitude of the Tafel slope.

The temperature T appears in the denominator of the exponential factors. A higher temperature reduces the sensitivity of the exponential factors to overpotential and increases the magnitude of the Tafel slope.

Thus,

$$i_0, \alpha, n, \text{ and } T \text{ control the kinetic response.}$$

3.5) Connection to the Nernst equation

The Nernst equation was introduced in Exercise 1 to describe how the cell potential depends on the reaction quotient. In that exercise, the cell potential was written as

$$E_{\text{cell}} = E_{\text{cell}}^{\ominus} - \frac{RT}{\nu F} \ln(Q). \quad (53)$$

At equilibrium, $Q = K$ and the galvanic cell no longer produces a net cell potential:

$$E_{\text{cell}} = 0. \quad (54)$$

Thus, the Nernst equation describes the equilibrium thermodynamic state of an electrochemical reaction. It determines the equilibrium potential E_{eq} .

The Butler–Volmer equation starts from this equilibrium potential and describes what happens when the electrode is driven away from it. This is done through the overpotential

$$\eta = E - E_{\text{eq}}. \quad (55)$$

Therefore,

Nernst gives equilibrium; Butler–Volmer gives the kinetic response away from equilibrium.

If $E = E_{\text{eq}}$, then $\eta = 0$ and the Butler–Volmer equation gives zero net current. If $E \neq E_{\text{eq}}$, then $\eta \neq 0$ and a net faradaic current flows.

3.6) Connection to electric-double-layer models

The electric-double-layer models were studied after the thermodynamic cell potential exercise. These models describe how the electrostatic potential and ion concentrations vary near the electrode surface.

For example, Gouy–Chapman describes a diffuse ionic layer, while Gouy–Chapman–Stern adds a compact Helmholtz layer. Finite-size models such as Bikerman-type models further modify the near-surface ion distributions.

The Butler–Volmer equation does not describe this interfacial structure explicitly. Instead, it uses an overpotential to describe the driving force for charge transfer. In a more complete electrochemical model, the local potential drop across the double layer can affect the effective overpotential at the reaction plane.

Thus, the different exercises describe complementary parts of an electrochemical interface:

- Exercise 1 describes equilibrium thermodynamics and the Nernst equation.
- Exercises 2–3 describe electric-double-layer structure.
- Exercise 4 describes charge-transfer kinetics through Butler–Volmer.

In this sense, a complete electrochemical model often needs all three ingredients: thermodynamics to set the equilibrium potential, double-layer models to describe interfacial electrostatic structure, and Butler–Volmer kinetics to describe the faradaic current generated by charge transfer.

Thus,

thermodynamics + double-layer structure + kinetics

are all needed for a complete electrochemical interface description.

References

- [1] O. Schmidt, “Exercise 4: Charge-transfer kinetics: Butler–volmer,” 2026. Problem sheet.